

LINEAR SEARCH

LINEAR SEARCH EXPLAINED

- The linear search algorithm searches sequentially through each element in an array.
- The algorithm tests each element in the array to see if it matches the search key.
- If the algorithm reaches the end of the array and the search key has not been found the user is informed that the search key is not present.
- If the algorithm finds a match in the array for the search key, it returns the index of that element.

LINEAR SEARCH USES

- A linear search can be used to:
 - look up a word in a dictionary
 - find a library book
 - look through a list of contacts
 - search an unordered list

EXAMPLE

25	4	8	12	9	61	3	54	10
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Search Key	61
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- Using the linear search algorithm, the algorithm first checks if 25 matches the search key. It does not, so the algorithm checks if 4 matches the search key. It does not.
- The algorithm continues moving sequentially through the array, checking 8, then 12 and then 9.
- There isn't a match, so the algorithm then checks 61. It is a match.
- The algorithm returns the index 5 which is the location of 61 in the array. Remember arrays are 0-based.

ANOTHER EXAMPLE

25	4	8	12	9	61	3	54	10
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Search Key	12
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- Using the linear search algorithm, the algorithm first checks if 25 matches the search key. It does not, so the algorithm checks if 4 matches the search key. It does not.
- The algorithm continues moving sequentially through the array, checking 8, then 12.
- 12 is a match.
- The algorithm returns the index 3 which is the location of 12 in the array. Remember arrays are 0-based.

LINEAR SEARCH IMPLEMENTATION

- The linearSearch method performs the linear search.
- Two parameters - the array to be searched and the search key
- A loop traverses the array
- Each element of the array is in turn compared to the search key
- If there is a match, return the index
- If there is no match, return -1

```
public static int linearSearch(int data[], int searchKey){  
    for(int i=0; i<data.length; i++){  
        if(data[i] == searchKey){  
            return i;  
        }  
    }  
    return -1;  
}
```